1D Numeric Arrays – Scenario-Based Programming Questions

Difficulty Level: Medium
Topics Covered: Array declaration, initialization, processing, searching and sorting

1. Student Performance Analysis (Array Processing)

A class has N students. The marks obtained by each student in a programming exam are stored in a 1D array.

Write a C program to:
1. Read the number of students and their marks.
2. Calculate and display:
   • Total marks of the class
   • Average mark
   • Highest mark scored
   • Number of students who scored above the average mark

Sample Input:
Enter number of students: 6
Marks:
78 65 92 45 88 70

Sample Output:
Total Marks = 438
Average = 73.00
Highest Mark = 92
Students above average = 3

Concepts Tested:
Array input/output, sum calculation, maximum element, conditional processing

2. Product Price Search System (Linear Search)

A supermarket maintains the prices of N products in an array. A customer wants to check whether a particular product price exists.

Write a C program to:
1. Store the prices of products in a 1D array.
2. Accept a price value entered by the customer.
3. Perform Linear Search to find whether the price exists.
4. Display the position of the product if found.

Sample Input:
Number of products: 7

Prices:
120 250 75 500 300 150 90
Search price: 300

Sample Output:
Product price found at position 5

If not found:
Product price not available

Concepts Tested:
Linear search, array traversal, flag variable

3. Employee Salary Ranking System (Sorting)

A company stores the monthly salaries of employees in an array.

Write a C program to:
1. Read the salaries of N employees.
2. Sort the salaries in descending order using Bubble Sort.
3. Display the sorted salaries.
4. Display the highest and lowest salary.

Sample Input:
Number of employees: 5
Salaries:
35000 52000 28000 45000 60000

Sample Output:
Salary Ranking:
60000
52000
45000
35000
28000

Highest Salary = 60000
Lowest Salary = 28000

Concepts Tested:
Bubble sort, swapping, maximum/minimum after sorting

4. Online Shopping Discount Analysis (Searching + Processing)

An online shopping website stores the prices of items purchased by a customer.

If a customer searches for a particular item price:
• If found, apply a 10% discount on that item.
• If not found, display "Item not available".

Write a C program to:
1. Store item prices in a 1D array.
2. Search for the given price using Linear Search.
3. Calculate and display the discounted price.

Sample Input:
Number of items: 6
Item prices:
800 1200 500 2500 1500 900
Search item price:
1500

Sample Output:
Item found at position 5
Original Price = 1500
Discount = 150
Final Price = 1350

Concepts Tested:
Searching, arithmetic processing, conditional statements

5. Hospital Patient Temperature Monitoring (Array + Sorting)

A hospital records the body temperatures of patients admitted during a day.

Write a C program to:
1. Read temperatures of N patients into an array.
2. Find:
   • Maximum temperature
   • Minimum temperature
   • Number of patients having fever (temperature >= 100°F)
3. Arrange the temperatures in ascending order using Bubble Sort.

Sample Input:
Number of patients: 7
Temperatures:
98 101 99 103 97 100 102

Sample Output:
Maximum Temperature = 103

Minimum Temperature = 97
Patients with fever = 4

Sorted Temperatures:
97 98 99 100 101 102 103

Concepts Tested:
Array processing, counting, conditional search, bubble sort

Recommended Practice Order

1. Student Performance Analysis – Basic array processing
2. Product Price Search – Linear search
3. Employee Salary Ranking – Sorting
4. Shopping Discount – Search and calculation
5. Hospital Monitoring – Multiple array operations and sorting